

Topology Strategies

1. Connectedness

1.1. Proving

- Cont image of connected space
- Intervals in $\mathbb{R}$
- Union of connected sets with nonempty intersection
- Path-connected

1.2. Disproving

- Find a separation
- Continuous map to a disconnected space
- Find a nontrivial clopen subset

2. Compactness

2.1. Proving

- Heine-Borel: in $\mathbb{R}^n$ compact $\Leftrightarrow$ closed + bounded
- Cont image of a compact space
- Closed subset of a compact space
- Sequential compactness

2.2. Disproving

- Find an open cover with no finite subcover
- Heine-Borel
- Sequence with no convergent subsequence

3. Homeomorphism

3.1. Proving

- Construct a homeomorphism (bicontinuous bijection)
- (a, b) is homeomorphic to $\mathbb{R}$
- Continuous bijection from compact space to Hausdorff space

3.2. Disproving

- Homeomorphic spaces share all topological invariants
 - Connectedness
 - Compactness
 - # of connected components
 - Removal of points
 - Local properties
 - Cardinality

4. Metrizable

4.1. Proving

- Construct a metric
- Embed into a metric space
- Smirnov-Nagata (T_3 with countable locally finite base)
- Urysohn (second countable and T_3)

4.2. Disproving

- Break one of the properties
 - First countable
 - T_3
 - Normal
 - Sequences characterize closure
 - Points are G_δ